INFLUENCE OF THE COBALT CONTENT IN HOT-PRESSED CEMENTED CARBIDES ON THE DEPOSITION OF LOW-PRESSURE DIAMOND LAYERS

R. Haubner, B. Lux

JOURNAL DE PHYSIQUE
Colloque C5, supplément au n°5, Tome 50, mai 1989

INFLUENCE OF THE COBALT CONTENT IN HOT-PRESSED CEMENTED CARBIDES ON THE DEPOSITION OF LOW-PRESSURE DIAMOND LAYERS

R. HAUBNER and B. LUX

Institute for Chemical Technology of Inorganic Materials, Technical University Vienna, A-1060 Vienna, Austria

Résumé - Il est d'un grand intérêt de déposer une couche de diamant sur des substrats de métaux durs utilisés pour la coupe de matériaux tels que les alliages Al-Si, les fontes, les bois, le bois dòenvie, les plastiques renforcés, etc.
L'obtention d'une couche continue et adhérente de diamant sur des alliages conventionnels de métaux durs contenant plus de 5 % en poids de cobalt s'est avérée, être difficile.

On a déposé du diamant sur des substrats de WC comprimés à chaud contenant 0; 0,3; 1; 2 et 3 % en poids de cobalt pour examiner l'influence possible de celuici.

Le dépôt a été effectué sur des surfaces polies par dépot en phase vapeur à l'aide d'un filament chaud et en utilisant comme mélanges gazeux H_2 + CH_4. Les couches déposées ont été examinées on microscope électronique à balayage et par diffractométrie des rayons X.

Il a été prouvé que le cobalt exerce une mauvaise influence à l'interface diamant/métaux durs. On n' a pu déposér des couches continues et adhérentes de diamant que sur des surfaces sans cobalt ou n'en contenant que peu.

Abstract - Diamond coatings deposited on cemented carbide substrates are of interest for high-performance cutting of materials such as Al-Si alloys, cast iron, woods, timber, fiber reinforced plastics, etc. The formation of a continuous and adherent diamond layer on conventional cemented carbide alloys containing more than 5 weight% Co binder proved however to be difficult.

Hot-pressed WC substrates with 0, 0.3, 1, 2 and 3 weight % Co were coated with diamond in order to examine the possible influence of Co. The deposition was carried out on polished surfaces using the hot filament assisted CVD method and H_2/CH_4 gas mixtures. The deposits were examined by scanning electron microscopy and X-ray diffraction techniques.

Co proved to exert a negative influence at the diamond/cemented carbide interface. Only on Co-free or low Co surfaces of cemented carbide substrates could continuous and adherent diamond layers be deposited.

1. INTRODUCTION

Diamond's high hardness as well as its chemical and wear resistance /1,2/ are of great interest for industrial applications and have led to its wide spread application for cutting, grinding, polishing, etc. /3/.

Although conventional natural and synthetic single- and polycrystalline diamonds are rather expensive, the low pressure diamond methods offer the possibility to produce specific tools for industrial purposes at much lower costs /4/. For several years this syntheses - operating at low pressure and under nonequilibrium conditions - have been under development /5-8/.

The method used in this work is based on hot filament assisted chemical vapour deposition /10-14/. Other methods are plasma activated CVD using microwave and radio-frequency plasma /15-17/. Diamond coatings on tools should have properties very close or better to those found in conventional single- and polycrystalline diamonds. Their application can partially substitute or improve PCD layers on cemented carbides and bulk PCD cutting tools/18/. Diamond deposition on cemented carbides however suffers still from poor adhesion in the presence of cobalt /19,20/, though during industrial cutting tests on special materials such as Al-Si alloys or presintered SiC, these coatings have already shown interesting performances /21-23/.

In this work the influence of different cobalt contents in cemented carbides on the growth of continuous diamond coatings was studied.

Article published online by EDP Sciences and available at http://dx.doi.org/10.1051/jphyscol:1989524

2. EXPERIMENTAL PROCEDURE

Pure WC single crystals produced by the Menstruum method were used as substrates for diamond deposition without any surface preparation.

Cemented carbides with 0, 0.3, 1, 2 and 3 wt.% Co were produced by hot pressing. WC (grain size 1.16 µm) and Co (1.50 µm) powders were mixed in a ball mill for 24 hours. The mixed powders were then subjected to a pressure of 1.5 atm and a temperature between 1700 and 1750° C in a 15 mm diameter graphit die for the few minutes necessary to complete the sintering process. After grinding with diamond followed by polishing with SiC they were coated with diamond.

The method for diamond deposits using the hot filament thermal CVD method /9-13/ and the specific reactor /14/ have already been described elsewhere. Table 1 shows the typical deposition parameters.

The distance between sample and tantalum filament (0.3 mm diameter, 15 turns) was 5 mm. After evacuating the reaction chamber, a H_2/CH_4 mixture was allowed to flow through and the chosen working pressure was set. For heating the substrates to the deposition temperature an electrical furnace was used.

The substrate temperature during the experiment was monitored by a Ni/NiCr thermocouple placed in contact with the lower part of the substrate. The substrate surface temperature was indirectly determined by extrapolation based on a number of temperature measurements carried out in earlier experiments /24/. The filament temperature was measured directly with an optical pyrometer.

The diamond deposition were examined mainly by scanning electron micrograph (SEM) and X-ray diffraction.

3. EXPERIMENTAL RESULTS

3.1 Diamond deposition on the WC single crystals

Deposition time: 5 hours

- A long triangular prismatic WC single crystal showed in the early growth stage layer deposition only along the top edges (Fig.1a).
  The isolated diamond crystals nucleated on the crystal faces reached a grain size of about 5 µm; some clusters were formed (Fig.1b).
  The largest diamond single crystals formed had a maximum size of about 10 µm (Fig. 1c) and exhibited a five-fold symmetry (Fig.1d).
- If partial layer formation occurred on the crystal facettes due to a locally higher nucleation density (Fig.2a), then the diamond grain size in the layers was 2-4 µm, but the isolated crystals reached sizes up to 10 µm (Fig.2b,c,d).

3.2 Diamond deposition on cemented carbides with different Co contents

The diamond coatings on the different cemented carbide samples were examined after 5 or 16 hours deposition time (deposition parameters are given in Tab.1).

Deposition at 10 torr and 0.3 vol% CH_4

Deposition time: 5 hours: Isolated well facetted diamond crystals were formed

* At Co contents between 0 and 0.3%, the diamond particle size was about 5 µm.
* At contents of 2 and 3 % in the cemented carbides, the diamond crystals were about 3 µm in size (Fig.3).

16 hours: Diamond layer formation was observed.

* At 0 and 0.3 % Co the layers were well facetted with a grain size between 8 and 10 µm.
* At 3% Co small layers of ballas-type diamond were formed (Fig.4).

Diamond deposition at 20 torr and 0.5 vol% CH_4

Deposition time: 5 hours: Layer formation was observed on all substrates containing between 0 and 3 wt% Co.

Co contents:

* 0 and 0.3% Co led to deposited diamonds which were well-facetted with crystals up to 5 µm in size (Fig.5a,b).
* 1% Co in the hard metal led to an unidentified phase between the diamond crystals (Fig.5c).
* 2 and 3% Co led to nucleation of new diamond crystals during the entire deposition time so that many small diamond crystals are formed between only a few large crystals (Fig.5d,e).

4. DISCUSSION

4.1 Diamond nucleation and growth

* WC single crystals

For the diamond deposition observed on the single crystal facettes of WC, no clear orientation relationship between WC and diamond can be detected with SEM because of the five-fold symmetry of diamond deposition. Only for a few diamond crystals could a relationship be assumed. A differentiation applying Kossel X-ray methods is in preparation /25/. However, epitaxy of diamond on WC is not as obvious as in case of diamond on cub-BN /26/.

Diamond growth on WC single crystals leads, within the applied pressure and CH_4 concentration range, to well-facetted diamond crystals, with diamond layers being formed at long deposition times.

* Cemented carbides

In the case of cemented carbides it is known that the chemical nature of the powder used for the polishing operation has a great influence on the first nucleation rate /27,28/.

After the identical substrate surface preparation (grinding with diamond followed by a SiC polish) an influence of the Co contents (0-3% Co) in the cemented carbide on the diamond nucleation number and growth rate was no longer observed. The Co concentration did however affect the diamond morphology.

10 torr and 0.3 vol% CH_4

If the diamond nucleation rate is low, after the formation of a few diamond nuclei no further diamond nucleation occurs. This confirms earlier experiments on metal surfaces showing that below a critical distance between two diamond crystals, which is a function of the supersaturation in the gas phase, a new nucleation is impossible /29/.

20 torr and 0.5 vol% CH_4 :

Because of the higher gas pressure and CH_4 concentration, increased diamond nucleation was observed. The greater number of nuclei led to a regular layer formation. This confirms the relationship between high diamond nucleation rate and good layer formation observed earlier during deposition of diamond on molybdenum /30/.

4.2 Influence on nucleation

R.Bichler /31/ reported that the diamond nucleation rate on cemented carbides without a prior diamond polishing treatment of the surface increases with increasing Co concentration (3-10% Co) (Fig.6a). At a Co concentration higher than 6% the diamond nucleation rate reached a minimum. A negative influence of Co on diamond nucleation at cemented carbide surfaces was reported by H.Matsubara and J.Kihara /32/. They obtained a graphitic carbon deposition on the Co phase, which was later overgrown by the diamond, leading to the poor adhesion of the diamond coatings. This is difficult to observe for the fine-grained cemented carbides which were used here. However the same basic principals still apply. Furthermore "surface preparation" by diamond polishing does not completely prevent this negative effect. Apparently such induced nucleation does not necessarily improve adhesion either /28/.

4.3 Influence on morphology

The diamond morphology is clearly influenced by the Co concentration. At low concentrations (0 to 0.3 % Co) and at short deposition times no nucleation effect or growth changes due to Co were observed. At a higher Co content (3%) branching due to surface nucleation occured and smaller crystals were grown on top of the larger crystals. Co diffusion to the layer surface leads to a secondary nucleation of new diamond crystals. The same influence of diffusing Co on other CVD-depositions leading to changes in the morphology of the Al_2O_3 coatings has already been investigated using Auger analysis /33/.

The adverse influence of Co on the diamond growth rate was also reported by R.Bichler /31/ in cemented carbides with Co contents ranging from 0.3 to 10 wt%. The deposition rate increased with Co content as shown in fig.6b. Increasing the Co concentration from 0 to 6 % caused a 50% drop in the deposition rate (Fig.6b).

5. CONCLUSION

The cobalt contained in a cemented carbide substrate has a significant influence on the morphology of the deposited diamond. Low Co concentrations (<3% Co) have little effect on diamond nucleation and growth rates. Above this concentration however the Co content becomes important.

These results indicate that during diamond deposition on cemented carbides a surface with only a minimum Co concentration allows good diamond deposition. This is particularly important for the industrial formation of diamond coatings on cutting tools. Using other deposition parameters with higher growth rates (filament temperature 2600°C) leads to even better diamond deposition. The carbon deposition occurs faster than the Co can diffuse to the surface so that only little Co can reach the surface to influence the deposition mechanism /32/.

Acknowledgements

The authors would like to thank the "Fonds zur Förderung der wissenschaftlichen Forschung", Project P6031, for financial support and the students S.Turnovszky and A.Tschinkowitz for their help with the experimental work. We also want to thank SANDVIK Coromant for delivery of the WC single-crystal substrates.

6. LITERATURE

1) J.E.Field: The Properties of Diamond, Academic Press, London 1979
2) N.Fujimori, T.Imai, A.Doi: Proc.9th Symp.on ISIAT'85, Tokyo, 237 (1985)
3) H.R.Meyer: Oberfläche, Heft 2, 50-58 (1971)
4) K.Recker: Z.Dt.Gemmol.Ges 1/2, 5, (1984)
5) B.V.Deryagin: New Scientist, p.230, Oct.30, 1969
6) B.V.Deryagin, D.V.Fedoseev, K.S.Uspenskoya: Russ.J.Phys.Chem. 47 (1)(1973)
7) J.C.Angus, H.A.Will, W.S.Stanko: J.Appl.Phys., 39 (6), 2915 (1968)
8) H.Schachner, B.Lux, K.Stjernberg, A.Thelin, H.Tippmann: German Patent DE 3522583 (1985)
9) S.Matsumoto, Y.Sato, M.Kamo, N.Setaka: Jpn.J.Appl.Phys., 21(4), L183 (1982)
10) S.Matsumoto, Y.Sato, M.Tsutsumi, N.Setaka: J.Mat.Sci. 17, 3106 (1982)
11) N.Kikuchi, T.Komatsu, H.Yamashita: Proc. 9th Symp.on ISIAT'85, Tokyo, 233 (1985)
12) N.Kikuchi, T.Komatsu, H.Yamashita: Proc. 10th Symp.on ISIAT'86, Tokyo, 359(1986)
13) Y.Hirose, Y.Terasawa: Jpn.J.Appl.Phys.,25(6),L519 (1986)
14) R.Bichler, R.Haubner, B.Lux: Proc.6th Euro CVD, Jerusalem, 413 (1987)
15) S.Matsumoto: ISPC Eindhoven (July 1985) Paper A-1-10

16) Y.Mitsuda, Y.Kojima, T.Yoshida, K.Akashi: J.Mat.Science 22, 1557-62 (1982)
17) R.Haubner, B.Lux: J.Ref.Hard Metals Vol.6, No.4, 210-15 (1987)
18) N.O-Otake, H.Tokura, M.Yoshikawa, C.F.Yang: 1st Int.Conf.New Diamond Science and Technology Oct.24-26, 1988; Keidanren Kaikan, Tokyo, Japan, P2-12, p.140
19) H.Suzuki, H.Matsubara, K.Shibuki: J.Jap.Soc.Powder a.Powder Metallurgy Vol.34, No.1, 22-25 (1986)
20) H.Suzuki, H.Matsubara, N.Horie: J.Jap.Soc.Powder a.Powder Metallurgy Vol.33, No.5, 262-269 (1986)
21) A.Ikegaya, M.Tobioka: Proc.6th Euro CVD,Jerusalem, 401 (1987)
22) M.Yagi, N.Horie, K.Shibuki: Proc.6th Euro CVD,Jerusalem, 428 (1987)
23) N.Kikuchi, T.Komatsu, H.Yoshimura: 3rd Int.Conf.Science Hard Materials Nov.8-13, 1987, Nassau, Bahamas
24) P.O.Joffreau, R.Bichler, R.Haubner, B.Lux: J.Ref.Hard Metals, Vol.7, No.2, 92-7 (1988)
25) H.J.Ullrich, R.Haubner, B.Lux: to be published
26) R.Haubner: Proc.Hard Materials Production 88, London Apr.11-13, 1988, 34 1-17
27) S.Yugo, T.Kimura: 1st Int.Conf. New Diamond Science and Technology Oct.24-26, 1988; Keidanren Kaikan, Tokyo, Japan, P2-20, p.130
28) X.X.Pan: Thesis TU-Vienna (to be published)
29) T.Anthony: MRS Fall Meeting, Nov. 30-Dec. 3, 1987, Boston
30) P.O.Joffreau, R.Haubner, B.Lux: J.Ref.Hard Metals, Vol.7, No.4 (1988) in press
31) R.Bichler: Thesis TU-Vienna 1988
32) H.Matsubara, J.Kihara: 1st Int.Conf. New Diamond Science and Technology, Oct.24-26, 1988, Keidanren Kaikan, Tokyo, Japan, pp.114-5
33) C.Colombier, B.Lux, A.Riahi, M.Puchhammer, H.Störi: Fresenius Z.Anal.Chem, 329, 355-60 (1987)

Table 1: Deposition parameters

filament temperature	: 2000 ± 10°C
substrate temperature	: 800 ± 10°C
substrate surface temp.	: 1075 ± 25°C
pressure	: 10 and 20 torr
gas composition	: 0.3 and 0.5 vol.% CH_4 in H_2
gas flow rate	: 1.8 sccm/min CH_4)
	550 sccm/min H_2
deposition time	: 5, 16 hrs
hydrogen	: 99.999 Vol.% H_2
methane	: 99.5 Vol.% CH_4

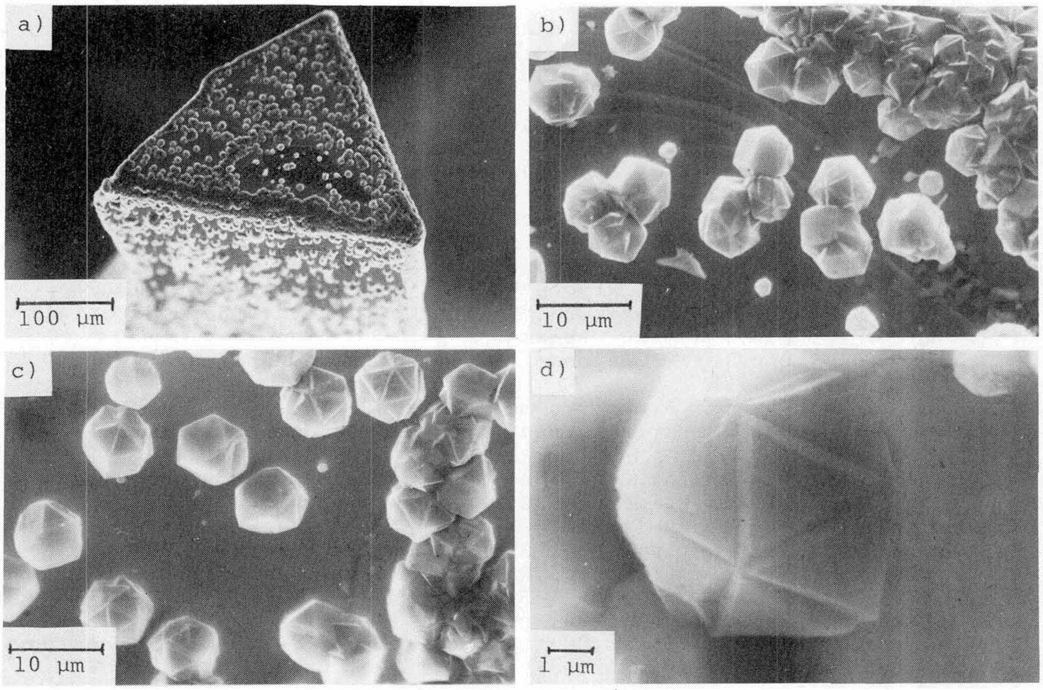

Fig.1: Diamond deposition on a triangular WC single crystal
a) overview
b, c) deposition on the top
d) five-fold symmetry

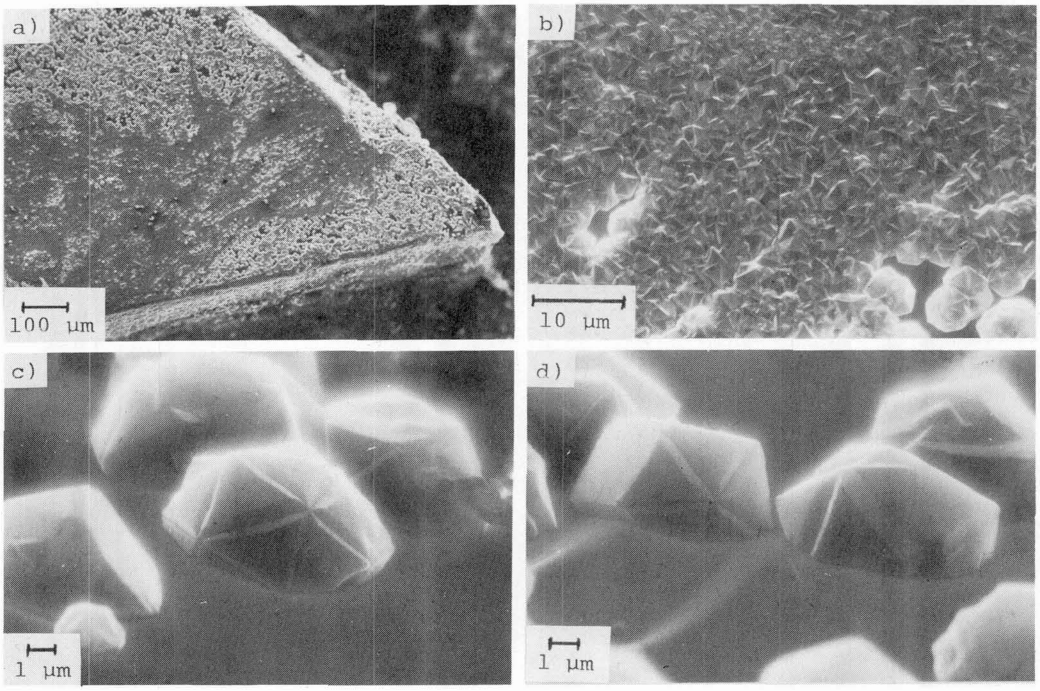

Fig.2: Diamond deposition on a large WC-single crystal facette
a) overview
b) layer formation
c, d) crystals with five-fold symmetry

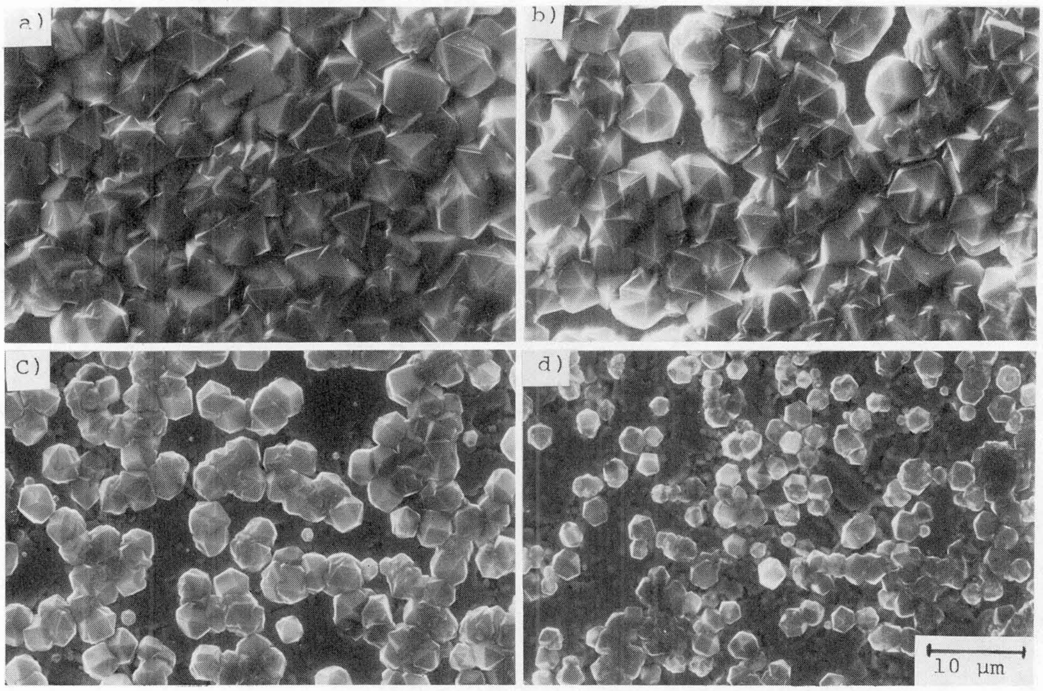

Fig.3: Deposition on cemented carbides containing different concentrations Co (5 hours dep.time, 10 torr, 0.3 vol% CH_4)
a) 0% Co b) 0.3% Co c) 2% Co d) 3% Co

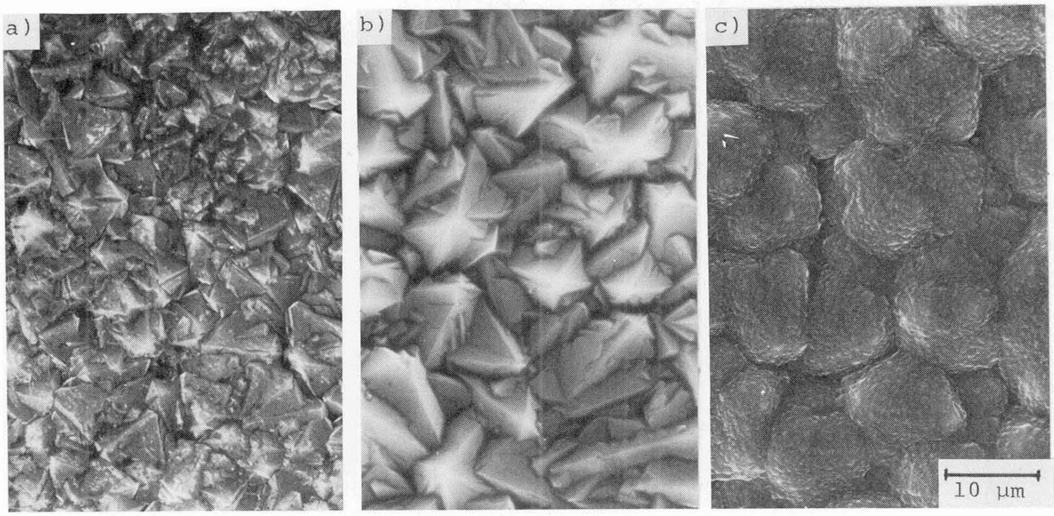

Fig.4: Layer formation on cemented carbide containing different Co concentrations (16 hours dep.time, 10 torr, 0.3 vol% CH_4)
a) 0% Co b) 0.3% Co c) 3% Co

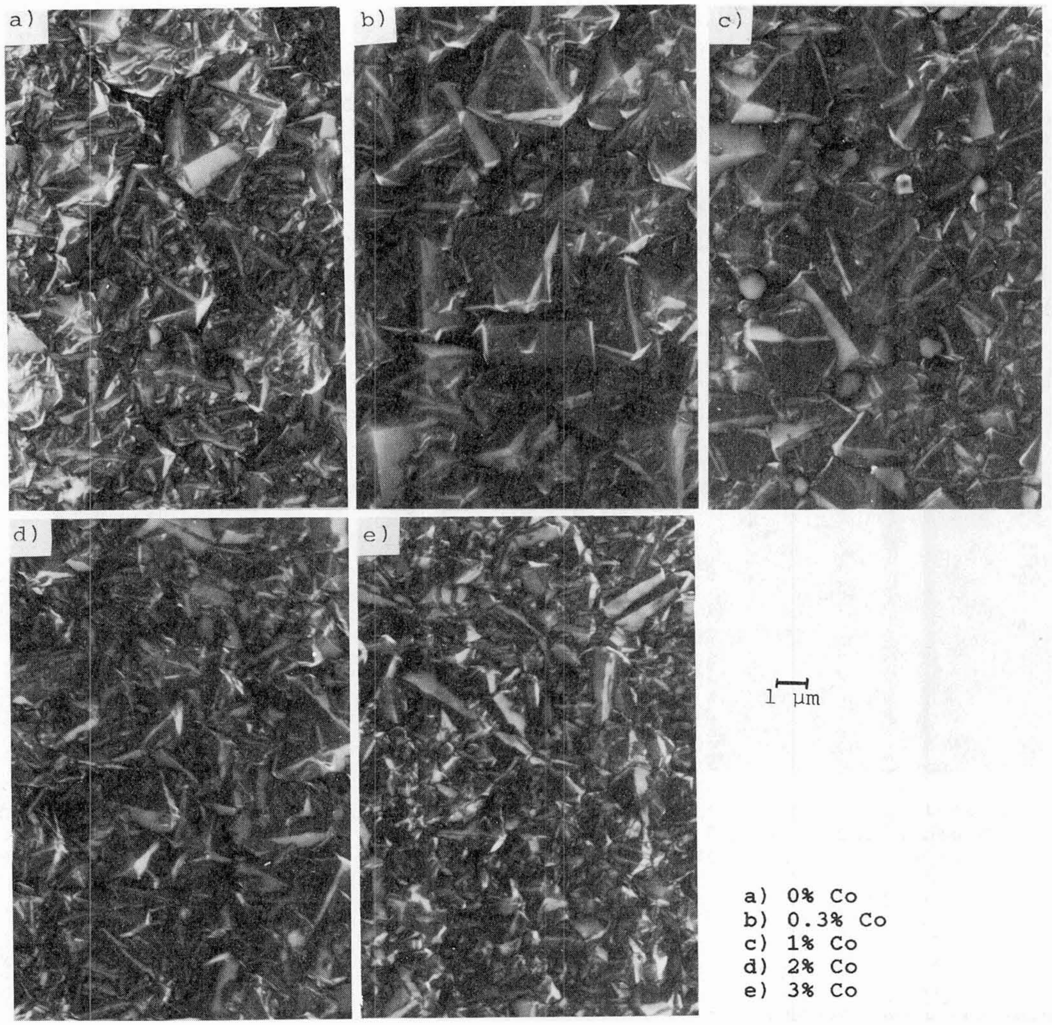

Fig.5: Changing of grain size in the layers due to different Co levels (5 hours dep.time, 20 torr, 0.5 vol% CH_4)

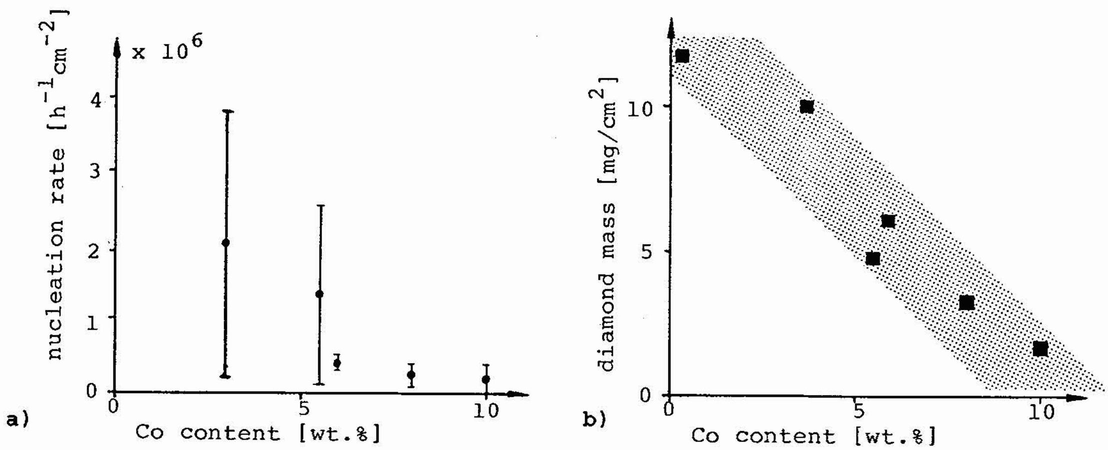

Fig.6: Influence of the Co concentration in cemented carbide on diamond deposition /31/ (800°C, 0.5 % CH_4)
a) diamond nucleation density b) diamond mass (16 h)